



**AI ASSISTED GRADING FRAMEWORK FOR THAI-LANGUAGE
WRITTEN EXAM QUESTIONS BASED ON LLM AND RULE-BASED
REASONING APPROACH**

BY

CHUTIPON TRIRATTANANURAK

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on June 6, 2025

Chairperson

(Assistant Professor Seksan Laitrakan, Ph.D.)

Member and Advisor

(Assistant Professor Sasiporn Usanavasin, Ph.D.)

Member and Co-advisor

(Chaianun Damrongrat, Ph.D.)

Member

(Professor Manabu Okumura, Ph.D.)

Member

(Firstname Lastname, Ph.D.)

Director

(Associate Professor Kriengsak Panuwatwanich, Ph.D.)

Dissertation Title	AI ASSISTED GRADING FRAMEWORK FOR THAI-LANGUAGE WRITTEN EXAM QUESTIONS BASED ON LLM AND RULE- BASED REASONING APPROACH
Author	Chutipon Trirattananurak
Degree	Master of Engineering (Artificial Intelligence and Internet of Things)
Faculty/University	Sirindhorn International Institute of Technology/ Thammasat University
Dissertation Advisor	Assistant Professor Sasiporn Usanavasin, Ph.D.
Dissertation Co-Advisor	Chaianun Damrongrat, Ph.D.
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ABSTRACT

Assessment plays a central role in education, providing insights into student comprehension, learning progress, and curriculum effectiveness. Written exams are particularly valuable as they capture higher-order cognitive skills such as reasoning, problem-solving, and expressive ability. However, manual grading of written responses is both time-consuming and prone to inconsistency due to factors such as evaluator fatigue, subjective interpretation, and variations in teaching assistant training. In the Thai context, these challenges are amplified by the language's complex grammar, tonal system, and context-dependent exceptions, which complicate evaluation and increase the likelihood of bias. Consequently, many educators rely heavily on multiple-choice examinations, which are easier to grade but insufficient for assessing deeper levels of understanding.

Recent advancements in Natural Language Processing (NLP), especially the development of Large Language Models (LLMs), provide new opportunities to address these challenges. LLMs offer sophisticated contextual analysis and semantic understanding that go beyond the limitations of traditional rule-based or keyword-based grading systems. However, their performance may still be unpredictable without structured guidance. This thesis proposes a hybrid AI-assisted grading framework that integrates the contextual reasoning

capabilities of LLMs with the transparency and consistency of rule-based reasoning. The framework is designed specifically for Thai-language written exam questions, aiming to reduce grading workload, enhance fairness, and support wider adoption of open-ended assessments in Thai education.

Through experimental evaluation, the framework demonstrates its ability to align closely with human grading while minimizing inconsistency. The results highlight its potential as a practical solution for teachers facing heavy workloads, as well as a step toward more equitable and comprehensive assessment methods in Thai education. This research contributes both a novel methodology for exam evaluation and empirical insights into the broader use of AI in educational contexts.

Keywords: Exam Evaluation, Natural Language Processing (NLP), Large Language Models (LLMs), Thai Language, Rule-Based Reasoning, Educational Assessment.

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LIST OF SYMBOLS/ABBREVIATIONS

Symbols	Terms
x	position
v	velocity
a	acceleration
t	time
F	force

Abbreviations	Terms
AI	Artificial Intelligence
LLMs	Large Language Models
NLP	Natural Language Processing
TA	Teaching Assistant
1G	First Generation
2G	Second Generation
3G	Third Generation
3GPP	3 rd Generation Partnership Project
4G	Fourth Generation
5G	Fifth Generation
5G-PPP	5G Infrastructure Public Private Partnership
ABS	Almost Blank Subframe
AC	Admission Control
AMPS	Advanced Mobile Phone System
AWGN	Additive White Gaussian Noise
BBU	Baseband Unit
BR	Best-Response
BS	Base Station
BYOD	Bring-Your-Own-Device
CA	Carrier Aggregation
CAPEX	Capital Expenditure
CBDSS	Cooperative Best-Response Distributed Subchannel Selection
CC	Control and Coordination
CCRM	Cooperative and Collaborative Resource Management
CDMA	Code Division Multiple Access
CESC	Cloud-Enabled Small Cell
CMM	Collaboration Management Module

CHAPTER 1

INTRODUCTION

Education relies heavily on assessment as a means of evaluating student learning, guiding curriculum development, and informing teaching practices. Among different assessment methods, written examinations stand out as a powerful tool for evaluating higher-order skills such as reasoning, problem-solving, and expressive ability. However, grading written exams poses significant challenges, particularly in contexts where linguistic complexity, large class sizes, and teacher workload combine to create barriers to fair and consistent evaluation. In Thailand, these challenges are even more pronounced due to the unique features of the Thai language and the increasing demand for efficiency in educational systems. This research addresses these issues by proposing an AI-assisted grading framework that leverages Large Language Models (LLMs) and rule-based reasoning to improve the evaluation of Thai written exam responses.

1.1 Background and Motivation

Examinations remain one of the most critical tools in educational assessment, serving as a benchmark for measuring students' comprehension, knowledge retention, and ability to apply concepts. Among different types of examinations, written exams are particularly valuable because they provide a deeper insight into students' higher-order cognitive abilities, such as critical reasoning, logical explanation, problem-solving, and language proficiency (Lee & Song, 2024). Unlike multiple-choice questions that primarily measure recognition and recall, written responses reveal students' thought processes, creativity, and capacity for structured argumentation.

Despite these advantages, written exam evaluation presents significant challenges. Traditional manual grading is time-consuming, resource-intensive, and often inconsistent. Human evaluators, including teachers and teaching assistants (TAs), are susceptible to fatigue, emotional states, and subjective interpretation, which may lead to inconsistency in grading outcomes (Månsson & Norgren, 2016). Even when grading rubrics are provided, variations in understanding and application among TAs may result in uneven assessment standards.

1.2 Challenges in the Thai Language Context

In the Thai educational context, the grading of written responses is further complicated by the unique linguistic and structural properties of the Thai language. Thai grammar is highly flexible, allowing for multiple valid sentence constructions. The language is tonal, meaning word meaning can change depending on pitch, and it contains numerous context-dependent exceptions (Tongsilp et al., 2024). Moreover, Thai writing omits spaces between words, which complicates segmentation and tokenization processes required for computational analysis.

These linguistic features make it particularly difficult for evaluators to interpret student responses accurately, especially when assessing reasoning, argumentation, or nuanced explanations. Teachers must balance linguistic expertise with an understanding of the subject matter and grading criteria. Given the open-ended nature of written responses, the risk of misinterpretation and grading bias is substantially higher compared to multiple-choice formats.

As a result, many Thai educators prefer multiple-choice examinations, which are quicker and easier to grade. This choice, however, comes at the expense of assessing higher-order skills. On average, teachers in Thailand manage classes of 100–400 students per subject while teaching 7–21 hours weekly and handling administrative, extracurricular, and pastoral responsibilities (Göksoy & Akdağ, 2014). Such workloads leave little room for the detailed evaluation of written work, leading to an overreliance on more easily graded exam types.

1.3 Limitations of Current Assessment Practices

Although multiple-choice exams are efficient, they fail to capture the richness of student reasoning, creativity, and expressive ability (Birenbaum, 1997; Scouller, 1998). Written exams, despite their resource-intensive nature, remain the most effective tool for evaluating comprehensive understanding and critical thinking (Gierl & Bulut, 2017; Gipps, 2011). This creates a tension between educational goals and practical constraints.

In recent years, efforts to explore the potential of LLMs in grading have included rubrics, peer evaluation, and partial automation through keyword-based scoring systems (Lu et al., 2021; Ridley & Husband, 2022). However, these approaches often fall short due to the complexities of natural language and the diversity of student expression. Rule-based systems, for instance, can detect certain patterns but struggle with semantic nuances, paraphrasing, and context-dependent meaning (Dikli, 2006; Shermis, 2014).

1.4 Role of Bloom’s Taxonomy in Assessment

Bloom’s Taxonomy provides a structured framework for categorizing cognitive skills, ranging from basic recall of facts to higher-order reasoning and critical thinking (Anderson & Krathwohl, 2001; Bloom et al., 1956). In the context of this research, Bloom’s Taxonomy serves two important purposes.

First, it provides a pedagogical lens for understanding the type of cognitive process each exam question is designed to assess. This ensures that grading criteria align with intended learning outcomes (Krathwohl, 2002).

Second, Bloom’s levels are used as a pre-screening mechanism for the AI-assisted framework. Each exam question is labelled with a Bloom’s category by expert teachers, and this label is then used to guide model selection. In practice, different types of questions (e.g., recall vs. analysis) may be better handled by different LLMs or combinations of LLM and rule-based reasoning (Stanny, 2016; Zohar & Dori, 2004).

By integrating Bloom’s Taxonomy into the framework, the system acknowledges both pedagogical validity and technical feasibility, linking educational theory with AI-based proof-of-concept evaluation.

1.5 Advances in Natural Language Processing (NLP)

Recent breakthroughs in Natural Language Processing (NLP) and the advent of Large Language Models (LLMs) have opened new possibilities for automated exam evaluation (Agarwal et al., 2021). LLMs demonstrate strong capabilities in tasks such as text classification, summarization, sentiment analysis, and automated evaluation across multiple languages. Unlike traditional rule-based methods, LLMs leverage deep learning and attention mechanisms to capture contextual meaning, making them suitable for handling open-ended and nuanced responses.

For the Thai language, the application of LLMs introduces unique opportunities and challenges. While most existing LLMs are trained primarily on English or other widely spoken languages, recent developments have included multilingual models capable of handling Thai to varying degrees. By integrating LLMs with rule-based reasoning, it may be possible to design a hybrid evaluation framework that combines the flexibility and contextual understanding of LLMs with the consistency and transparency of structured rules.

1.6 Research Motivation and Contribution

The motivation behind this research is twofold:

1. To reduce the grading burden on Thai teachers while improving fairness and consistency in written exam evaluation.
2. To promote the wider adoption of open-ended questions in Thai education, thereby encouraging the development of critical reasoning, creativity, and expressive skills in students.

This thesis aims to develop an AI-assisted grading framework for Thai-language written exam questions. By leveraging LLMs in combination with rule-based reasoning, the framework is designed to enhance accuracy, consistency, and efficiency in grading, while maintaining interpretability for educators.

The expected contributions of this research are as follows:

- A comprehensive analysis of challenges in grading Thai written responses.
- A proof-of-concept framework combining LLMs and rule-based reasoning.
- Empirical evaluation of the framework's performance compared to human graders.
- Insights into the broader implications for educational assessment and curriculum design.

1.7 Research Objectives

The objectives of this research are as follows:

1. To investigate the challenges and limitations in grading Thai written exam responses.
2. To design a proof-of-concept AI-assisted grading framework that integrates LLMs with rule-based reasoning.
3. To evaluate the framework's effectiveness against human grading in terms of fairness, consistency, and interpretability.
4. To provide insights into how such frameworks may support, rather than replace, teachers in Thai educational contexts.

1.8 Scope of the Study

This study is limited to the following scope:

1. Focuses on Thai-language written exam questions at the secondary school level.

2. Examines only short-answer and essay-type responses (0–3 scoring scale).
3. Uses a proof-of-concept approach with selected LLMs and teacher-designed rubrics.
4. Each exam question is pre-classified into Bloom’s Taxonomy levels by expert teachers, and these labels are used to guide model selection within the framework.
5. The framework is evaluated on fairness, consistency, and interpretability, but not on large-scale deployment or integration into actual school systems.

1.9 Structure of the Thesis

The remainder of this thesis is organized as follows:

- **Chapter 2 – Literature Review:** Reviews existing work on exam evaluation, NLP in education, and applications of LLMs in AI-assisted grading frameworks.
- **Chapter 3 – Methodology:** Describes the proposed framework, including data collection, preprocessing, model selection, and rule-based integration.
- **Chapter 4 – Result and Discussion:** Presents the experimental findings, discusses the strengths and limitations of the framework, and compares it with baseline methods.
- **Chapter 5 – Conclusion and Future Work:** Summarizes key findings, highlights contributions, and suggests directions for future research.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews prior work on automated exam evaluation, the use of large language models (LLMs) in education, and cognitive assessment frameworks. The thesis does not aim to build a new grading system but to test how reliable LLM-generated results are when compared with teacher evaluations. The review focuses on three areas: (1) applications of LLMs in educational assessment, (2) their reasoning and text-understanding abilities, and (3) approaches for validating AI results against human judgments. The chapter closes with Bloom’s taxonomy as a framework for aligning both AI and teacher assessments.

2.1 The Integration of LLMs in Educational Assessment

With the rapid development of LLMs, researchers have begun to explore their role in automating feedback and assessment, especially for open-ended and subjective tasks. Holmes and Tuomi (2022) argue that LLMs can enhance large-scale education by providing consistent, immediate feedback while simultaneously raising concerns regarding cultural bias and fairness in underrepresented languages. This issue is particularly relevant for Thai, where available training data is limited compared to high-resource languages such as English or Chinese.

Lan and Chen (2024) discusses the dual role of LLMs in education: on one hand, they support personalized learning, offering tailored feedback and explanations to students; on the other hand, they may encourage academic dishonesty by enabling students to outsource thinking tasks to AI. Similarly, Ryznar (2023) warns of potential misuse in assessment contexts, recommending redesigns of exam formats and the use of time-limited conditions to prevent LLM-driven plagiarism or unfair advantages.

These studies collectively highlight both the promise and pitfalls of adopting LLMs in educational systems. While they show potential to alleviate teacher workload and improve grading efficiency, they also underscore the importance of trustworthy, transparent, and context-sensitive systems.

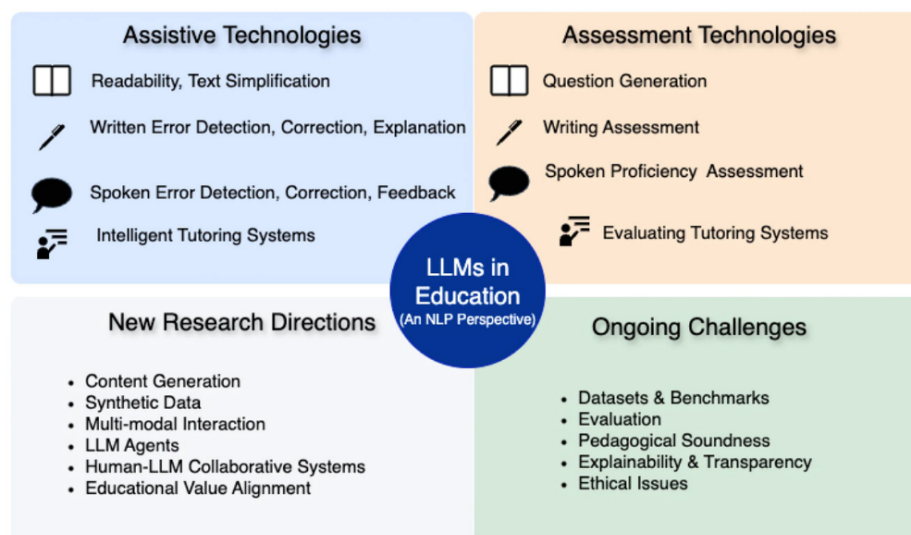


Figure 2.1 Opportunities and Challenges of LLMs in Educational Assessment. Source: Vajjala et al. (2025)

2.2 Leveraging LLMs for Text Understanding and Cognitive Inference

While LLMs are often praised for their text generation capabilities, their performance in logical reasoning and semantic inference is less consistent. Jang and Lukasiewicz (2023) demonstrated that GPT-4 performs well in syntactic and linguistic tasks, but struggles with negation, contradiction, and inference-based reasoning. This suggests that while LLMs capture surface-level patterns, deeper cognitive alignment may require additional strategies.

Lin et al. (2024) explored model fine-tuning, showing that applying lightweight methods such as LoRA to models like LLaMA-3 allows specialization in tasks such as sentiment intensity regression. Although fine-tuning is outside the scope of this thesis, their findings emphasize the need to align models with specific cognitive tasks rather than expecting universal performance from a single model.

In this study, Bloom's taxonomy is adopted as a classification lens: mapping student responses to different cognitive levels and selecting the most appropriate evaluation strategy (rule-based or LLM-based). This avoids the pitfalls of a one-size-fits-all method and ensures better alignment between assessment goals and AI capabilities.

2.3 LLM Ensembles and Model Selection Strategies

Since no single LLM excels universally across all tasks, ensemble strategies have gained attention as a way to mitigate weaknesses. Sakai and Lam (2025) introduced QUAD-LLM-MLTC, an ensemble framework for healthcare multi-label classification, which showed significant improvements in semantic accuracy compared to single-model baselines. Simi-

larly, Tekin et al. (2024) proposed LLM-TOPLA, which clusters model outputs to enhance robustness and efficiency in processing.

These works demonstrate that strategic model selection or combination of models can enhance reliability in real-world applications. Inspired by this, the current study adopts a hybrid strategy: selecting the best-performing LLM for each Bloom’s taxonomy category while integrating rule-based reasoning for tasks that lend themselves to deterministic evaluation.

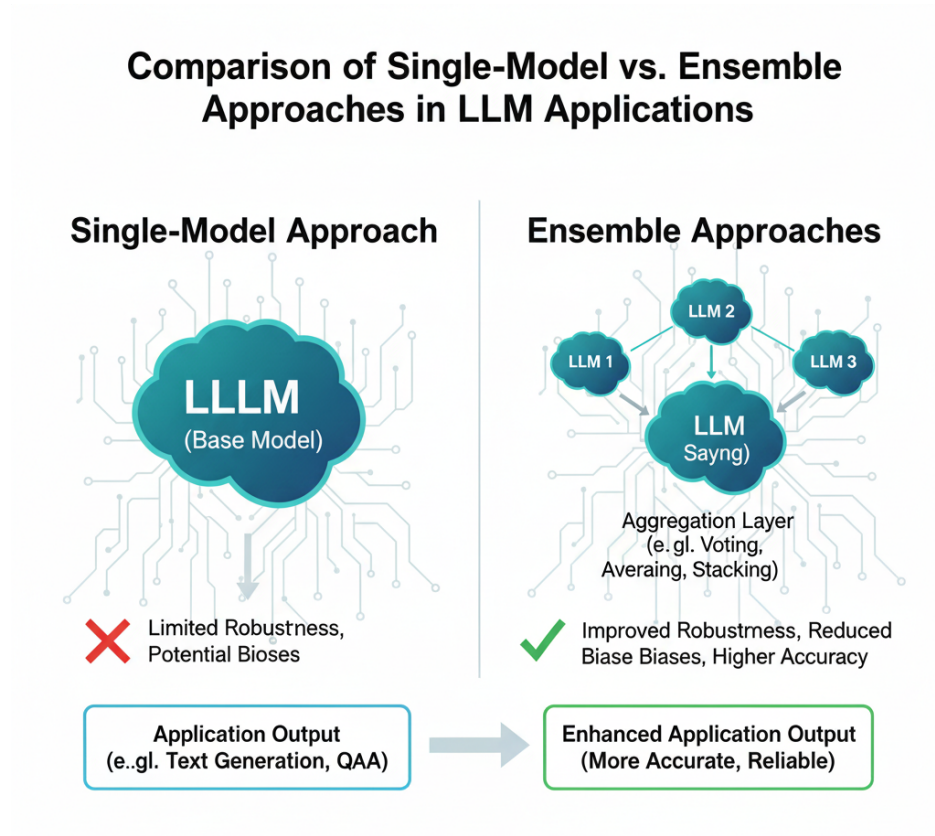


Figure 2.2 Comparison of Single-Model vs. Ensemble Approaches in LLM Applications

2.4 Bloom’s Taxonomy in Educational AI and Cognitive Classification

Bloom’s taxonomy provides a structured theoretical foundation for aligning assessment with cognitive skill levels. It divides learning objectives into six hierarchical categories:

1. **Remembering** – recalling factual knowledge.
2. **Understanding** – explaining ideas or concepts.
3. **Applying** – using knowledge in new contexts.
4. **Analyzing** – breaking down information into components.

5. **Evaluating** – making judgments or justifying decisions.

6. **Creating** – producing new or original work.

This taxonomy has been widely adopted in educational AI research. Abduljabbar and Omar (2015) applied machine learning to classify questions into Bloom’s levels, showing its utility for structuring automated evaluation. Rodrigo and Penas (2015) emphasized the importance of inter-rater agreement and gold-standard datasets, both critical for training reliable assessment systems.

More recent studies explore hybrid approaches. Dodia et al. (2023) combined rule-based methods with machine learning to grade short-answer responses, while Hubbard et al. (2017) demonstrated that question formats (e.g., multiple-true-false vs. free-response) influence the depth of student reasoning captured.

These findings support the argument that Bloom’s taxonomy is not only a theoretical tool but also a practical framework for guiding AI-supported assessment. By aligning AI evaluation methods with cognitive levels, systems can achieve greater accuracy, transparency, and interpretability.

Table 2.1 Six Categories of the Cognitive Domain Based on the Revised Bloom’s Taxonomy (2001) With Illustrative Examples

Category	Sample Keywords	Sample Behaviours / Sample Questions
Remembering	Define, list, name, recall, recognize	Students are able to define the 6 levels of Revised Bloom’s Taxonomy.
	Knowledge:	Recall basic facts and terms.
	Sample Questions:	Define each level of Revised Bloom’s Taxonomy.
Understanding	Classify, describe, discuss, explain, summarize	Students can explain the purpose of the Revised Bloom’s Taxonomy.
	Knowledge:	Explain relationships among concepts.
	Sample Questions:	Explain the purpose and importance of each level.
Applying	Demonstrate, implement, solve, use, execute	Students write an instructional objective for a method in the program.

Category	Sample Keywords	Sample Behaviours / Sample Questions
Analyzing	Knowledge:	Apply knowledge to new situations.
	Sample Questions:	Demonstrate the structure of each level in a program.
Evaluating	Analyze, compare, contrast, differentiate, examine	Students compare and contrast cognitive and affective domains.
	Knowledge:	Break concepts into parts and examine relationships.
	Sample Questions:	List advantages/disadvantages of using ArrayList vs array.
Creating	Assess, critique, defend, judge, validate	Students judge the effectiveness of Revised Bloom's Taxonomy.
	Knowledge:	Make judgments based on criteria and standards.
	Sample Questions:	Justify the concept of inheritance and give code examples.
Evaluating	Assess, critique, defend, judge, validate	Students judge the effectiveness of Revised Bloom's Taxonomy.
	Knowledge:	Make judgments based on criteria and standards.
	Sample Questions:	Justify the concept of inheritance and give code examples.
Creating	Assemble, construct, design, formulate, invent	Students design a classification method for educational objectives.
	Knowledge:	Combine elements to form a coherent whole or original product.
	Sample Questions:	Create a new method integrating cognitive, affective, and psychomotor domains.

2.5 Summary of Literature Review

The reviewed literature highlights three key insights relevant to this thesis:

- **LLMs in Education:** They hold significant potential for automating grading and feedback, but face challenges of fairness, cultural bias, and academic integrity.
- **Text Understanding and Reasoning:** LLMs demonstrate strengths in syntax and semantics but remain inconsistent in logical reasoning, supporting the need for hybrid frameworks.

- **Ensemble and Cognitive Alignment:** Combining multiple LLMs or matching models to cognitive levels, guided by Bloom's taxonomy, enhances reliability and interpretability.

Building upon these insights, this thesis proposes a hybrid AI-assisted grading framework tailored to Thai written exams. The framework integrates LLMs with rule-based reasoning, structured under Bloom's taxonomy, to ensure fair, consistent, and pedagogically aligned evaluation.

CHAPTER 3

METHODOLOGY

This study evaluates the effectiveness of Large Language Models (LLMs) in grading Thai-language written questions, aligning results with Bloom’s Taxonomy and expert human judgment. The process comprises five phases: question classification, dataset preparation, system setup, comparative evaluation, and data analysis.

3.1 Classifying Exam Questions based on Bloom’s Taxonomy

All of our written questions were categorized into Bloom’s six cognitive levels by two experienced Thai educators. Independent classifications were reconciled through consensus, ensuring pedagogical alignment for subsequent model evaluation.

3.2 Dataset Collection and Preparation

The exam questions are divided into two sets:

1. Rubric-Based Set: Rubrics were created per question using ChatGPT-extracted keywords (language accuracy, communication skills), verified by expert teachers.

“โปรดระบุคีย์เวิร์ดหลัก 3 คำจากคำถามต่อไปนี้
เพื่อใช้เป็นเกณฑ์ในการประเมินและตรวจคำตอบ”

Figure 3.1 Sample prompt used to extract key rubric keywords from Thai exam questions via ChatGPT (Please specify three main keywords from the following question to be used as criteria for evaluating and scoring the answer).

2. Non-Rubric Set: This set skips the rubric generation step to observe the LLMs’ raw interpretive grading ability without explicit criteria.

Five LLMs (ChatGPT, Typhoon, Gemini, Grok, Claude) generated 10 sample student responses per question, each scored 0–3 with explanations. Representative answers for each score level were refined by professional teachers and stored in Excel for contextual retrieval during grading. The prompt used was:

“โปรดสร้างตัวอย่างคำตอบของนักเรียนจำนวน 10 รูปแบบ สำหรับคำถามหนึ่งข้อ โดยแต่ละคำตอบควรได้รับคะแนนตั้งแต่ 0 ถึง 3 จากคะแนนเต็ม 3 คะแนน (อย่างหลากหลาย)

ให้แสดงผลลัพธ์ในรูปแบบ plain text โดยแต่ละตัวอย่างต้องประกอบด้วย:

1. คำตอบของนักเรียน (แสดงให้ชัดเจน)
2. คะแนนที่ให้ (ระบุเป็นตัวเลข 0–3)
3. คำอธิบายเหตุผลที่ให้คะแนนนั้น (อธิบายให้ชัดเจน ไม่ใช่ bullet หรือหัวข้อย่อหน้า)”

The Exam Question
The Rubric (Non-Rubric will skip this)

Figure 3.2 Prompt used to generate student answers and scores from LLMs (Please create 10 variations of student answers for a single question, with each answer receiving a score from 0 to 3 out of a total of 3 points (in a diverse manner). The results should be presented in plain text format, with each example consisting of: 1. The student’s answer (clearly displayed), 2. The assigned score (indicated as a number from 0 to 3), 3. A clear explanation of the rationale behind the score (explained clearly, without using bullet points or headings).)

CHAPTER 4

RESULT AND DISCUSSION

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 a

5.1.1 Subsection 1

Sirindhorn International Institute of Technology (SIIT) offers three master of engineering programs, namely, Master of Engineering Program in Engineering Technology, Master of Engineering Program in Information and Communication Technology for Embedded Systems, and Master of Engineering Program in Logistics and Supply Chain Systems Engineering (Bocchini & Frache, 2013; Smith, 2005).

5.1.1.1 Sub-Subsection 1

Prokop et al. (2005) demonstrated Sirindhorn International Institute of Technology (SIIT) offers three master of engineering programs, namely, Master of Engineering Program in Engineering Technology, Master of Engineering Program in Information and Communication Technology for Embedded Systems, and Master of Engineering Program in Logistics and Supply Chain Systems Engineering.

5.1.1.2 Sub-Subsection 2

Montanari et al. (2003) found Sirindhorn International Institute of Technology (SIIT) offers three master of engineering programs, namely, Master of Engineering Program in Engineering Technology, Master of Engineering Program in Information and Communication Technology for Embedded Systems, and Master of Engineering Program in Logistics and Supply Chain Systems Engineering.

Sirindhorn International Institute of Technology (SIIT) offers three master of engineering programs, namely, Master of Engineering Program in Engineering Technology, Master of Engineering Program in Information and Communication Technology for Embedded Systems, and Master of Engineering Program in Logistics and Supply Chain Systems Engineering.

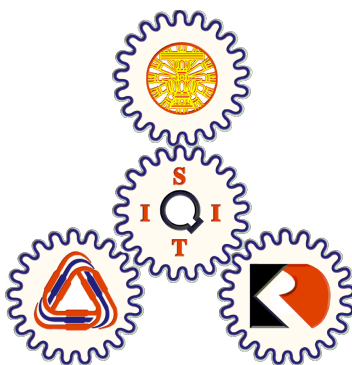
5.2 Latex Editor: Overleaf and TexMaker

You can do latex online via Overleaf or locally via programs such as Texmaker. Overleaf has a nice instruction for many things about Latex such as learning Latex in 30 minutes. See <https://www.overleaf.com/learn>.

5.3 Citation

5.3.1 Image and figure

This institute has two logo as shown in Figure 5.1. The SIIT and TU logos are shown in Figure 5.1a and Figure 5.1b, respectively.



(a) SIIT Logo



(b) TU Logo

Figure 5.1 Our institute logos

5.3.2 Reference

If you cite the related work within the sentence, the citation can be done by Bocchini and Frache (2013). On the other hand, this technique (Bocchini & Frache, 2013) is used when citation is outside the sentence.

Here are more examples: Smith (2005) and list of citations, Chen and Winterbone (2014), Garlotta (2001), and Prokop et al. (2005).

To organize our reference, we put the reference items in the file `thesis_ref_new.bib`. You may edit the bib file in many ways, Using a bibliography program like "jabref" (<https://www.jabref.org/>) is convenient. It works for all operating systems. There are other more recent bibliography softwares like Mendeley, EndNote, Zotero.

Jabref can import citation in many formats.

Here is example of reference within the document. The literature review is in Chapter 2.

5.3.3 Biblatex and Biber

For the citation and reference, we follow the APA style (<https://libguides.murdoch.edu.au/APA/text>). We use Biblatex and Biber instead of Bibtex. We need to configure our Latex editor, TexMaker, to use biber instead of bibtex. To do this we need to go "Options", then "Configure Texmaker" and change in the field Bib(la)tex from "bibtex %.aux" to "biber %.bcf" as shown in Figure 5.2. For more info, see <https://tex.stackexchange.com/questions/154751/biblatex-with-biber-configuring-my-editor-to-avoid-undefined-citations>. The link also shows how to configure TeXstudio, which is another latex editor.

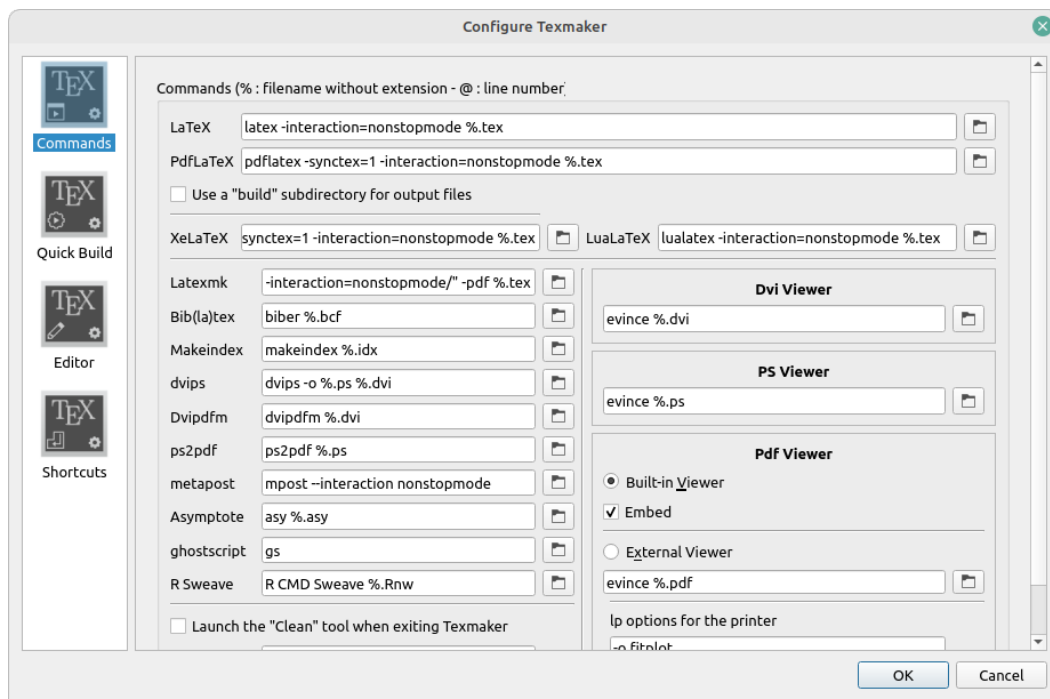


Figure 5.2 Texmaker Configuration to use Biblatex

5.3.4 URL Reference

Here is a clickable url link: https://www.overleaf.com/learn/latex/Creating_a_document_in_LaTeX. This is a beginner tutorial on using Latex. Overleaf is an online latex editor. You don't need to install anything to use latex there.

5.4 Equations

You can write an equation in the sentence as $x^2 + 1$ or in a separate sentence like

$$x = \sum_{n=0}^N z^n = \oint_C \vec{F} \cdot \vec{ds} = \frac{(\sqrt{x})^5}{\sin(x + \theta)} \quad (5.1)$$

We can refer to this equation as Equation (5.1). Note that SIIT requires equations to be aligned to the left margin. This is done by passing the option *fleqn* to the *amsmath* package. However, the equations still not aligned perfectly to the left margin. Hope this is ok with SIIT.

As we can see from Equation (5.1), the equation can be rewritten as (now we use the `\beq` and `\eeq` macros):

$$x = \frac{\partial u}{\partial t} = \nabla \cdot \vec{\mathbf{E}} = \begin{bmatrix} 1 & 2 & 3 \\ x & b & c \end{bmatrix} \quad (5.2)$$

where I've used several macros defined in `thesis_macros.tex`.

Equation (5.1) together with Equation (5.2) can be combined into... The case equation

$$\alpha(n) = \begin{cases} 0, & \text{if } n \text{ odd} \\ a_1 + a_{12}^{\ln 2}, & \text{otherwise} \end{cases} \quad (5.3)$$

We can write multi-line equation using "align" and &

$$2x - y = 8 \quad (5.4)$$

$$5x + 2y = 18 \quad (5.5)$$

We can remove the equation numbering in the first part of the equation with `\nonumber` or equivalently the macro `\nn`:

$$\begin{aligned} x &= z + 3 \\ &= y - z \end{aligned} \quad (5.6)$$

5.5 Tables

An example of a simple table is shown in Table 5.1. Note by default the table usually appears at top of a page. SIIT requires that the table must be center aligned, while the caption is left-aligned.

A multicolumn table is shown in Table 5.2. A multi-column multi-row table is shown in Table 5.3, which is generated automatically from the website <https://www.tablesgenerator.com/>. We make it appear directly below by putting [h] for "here".

Table 5.1 Simple table

Item	Value	Comment
1	2	left-aligned entry 555
2	x^2	oh! Equation in table!

Table 5.2 Multi-column table

Multi-column	
X	X
y	y

5.6 Figures

Figure 5.3 shows a simple figure.

For convenience, a macro `\bfigl` is defined. You can just pass the ratio of the image size relative to the paragraph width, the file name, and the caption:

**Figure 5.4** TU Logo

The macro is defined in `thesis_macros.tex` as

```
\newcommand{\bfigl}[3]{
  \begin{ figure }[H]
  \ centering
  \ includegraphics [width=#1\textwidth ]{#2}
  \ caption {#3}
```

Table 5.3 Multi-column, multi-row table

Item	SIIT			
	Rangsit		BKD	
1	2	3	4	5
6	7	8		

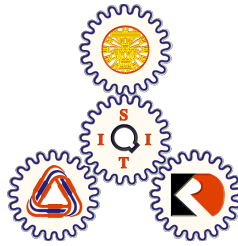


Figure 5.3 SIIT Logo

```
\label {#2}
\end{figure }
}
```

Other than convenience, we can change the macro easily. For example, if I want the figure to be left aligned, I just remove `\centering` in the macro definition.

You can also put `.png` into the file name. I set the macro such that the label is also the filename. Hence, the figure is Figure 5.4.

We can put multiple images into one figure as shown in Figure 5.1.

5.7 Pseudocode and Code Listing

Pseudocode can be written using the *algorithmicx* package <https://mirrors.mit.edu/CTAN/macros/latex/contrib/algorithmicx/algorithmicx.pdf>,

Algorithm 1 Euclid's algorithm

1: procedure EUCLID(a, b)	▷ The g.c.d. of a and b
2: $r \leftarrow a \bmod b$	
3: while $r \neq 0$ do	▷ We have the answer if r is 0
4: $a \leftarrow b$	
5: $b \leftarrow r$	
6: $r \leftarrow a \bmod b$	
7: end while	
8: return b	▷ The gcd is b
9: end procedure	

We can include codes like Python or C using *lstlisting* package: see https://www.overleaf.com/learn/latex/Code_listing. It shows even how to import codes from a file. This means we can just modify the codes by modifying at the source file and recompile Latex file.

5.8 Presentation

We can also create nice presentation slides with LaTeX. This is convenient since we already have our formulas in Latex. Examples of document classes are Beamer and Powerdot. See Overleaf tutorial for more into and online examples.

We can also embed Latex equations into Microsoft Powerpoint too. One way to do that is via *iguana*: <https://www.jonathanleroux.org/software/iguanatex/>.

5.9 Misc

You can put a box around a text, like creating a note:

Note 5.1. *This is the first note. The number runs automatically*

We can also hilighted words in **words** or **words** using \hili macro.

Here is another note:

Note 5.2. *This is the second note with equation*

$$x^2 + 1 \quad (5.7)$$

5.10 Review 1

Figure 5.5 shows analog voltage source (Riaz et al., 2022). The relation of voltage and current is shown in Equation (5.8). The Thammasart Logo is shown in Figure 5.4.



Figure 5.5 An analog voltage source.

$$V = IR \quad (5.8)$$

Note 5.3. *This is the third note.*

Now we see that all the auto numberings work nicely. They start with the chapter number 2.

5.11 Last Notes

1. Macros are wonderful: they can save you time typing. For example, if you happen to type electromagnetics a lot, it's wise to define a macro:

```
\newcommand{\em}{Electromagnetics}
```

The macros in `thesis_macros.tex` may give you some ideas.

2. When you write your thesis, you should regularly backup your files either to an external drive (USB, external HD) or cloud drive. You can also use online backup like svn or git.
3. The chapters are in separate files, as you can see from the `thesis_siit_temptate_250213.tex` file:

```
\include{chapter_01}
\include{chapter_02}
\include{chapter_03}
\include{chapter_04}
\include{chapter_05}
```

If you have large files, it is a good idea to only include the chapter you are working on by comment out the include of other files.

4. The file `thesis_siit_temptate_250213.tex` is the main document where all packages are included and macros are defined. In Texmaker, you should open this file and go to Options then "Define Current Document as 'Master Document'". If you use Overleaf, follow the instruction in https://www.overleaf.com/learn/how-to/Set_Main_Document.
5. Don't overwork. Exercise and rest sufficiently. Have fun writing.

REFERENCES

- Abduljabbar, D. A., & Omar, N. (2015). Exam questions classification based on bloom's taxonomy cognitive level using classifiers combination. *Journal of Theoretical and Applied Information Technology*, 78, 447–455.
- Agarwal, M., Kalia, R., Bahel, V., & Thomas, A. (2021). Autoeval: A nlp approach for automatic test evaluation system. *2021 IEEE 4th International Conference on Computing, Power and Communication Technologies (GUCON)*, 1–6. <https://doi.org/10.1109/gucon50781.2021.9573769>
- Anderson, L. W., & Krathwohl, D. R. (2001). *A taxonomy for learning, teaching, and assessing: A revision of bloom's taxonomy of educational objectives*. Longman.
- Birenbaum, M. (1997). Alternative assessment strategies: What do we know and what should we do? *Evaluation in Education*, 11(1), 29–44.
- Bloom, B. S., Engelhart, M. D., Furst, E. J., Hill, W. H., & Krathwohl, D. R. (1956). *Taxonomy of educational objectives: The classification of educational goals, handbook i: Cognitive domain*. David McKay Company.
- Bocchini, S., & Frache, A. (2013). Comparative study of filler influence on polylactide photooxidation. *Express Polym. Lett*, 7(5), 431–442.
- Chen, H., & Winterbone, D. (2014). A one-dimensional performance model for turbocharger turbine under pulsating inlet condition. *11th International Conference on Turbochargers and Turbocharging*, 113–123.
- Dikli, S. (2006). An overview of automated scoring of essays. *The Journal of Technology, Learning and Assessment*, 5(1).
- Dodia, S., Spoorthy, V., & Chandak, T. (2023). Machine learning-based automated system for subjective answer evaluation. *2023 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, 1–6. <https://doi.org/10.1109/conecct57959.2023.10234818>
- Garlotta, D. (2001). A literature review of Poly (lactic acid). *Journal of Polymers and the Environment*, 9(2), 63–84.
- Gierl, M. J., & Bulut, O. (2017). Automated essay scoring and the future of educational assessment in medical education. *Medical Education*, 51(4), 365–372.
- Gipps, C. (2011). *Beyond testing: Towards a theory of educational assessment*. Routledge.

- Göksoy, S., & Akdağ, Ş. K. (2014). Primary and secondary school teachers' perceptions of workload. *Creative Education*, 5(11), 877–885. <https://doi.org/10.4236/ce.2014.511101>
- Holmes, W., & Tuomi, I. (2022). State of the art and practice in ai in education. *European Journal of Education*, 57(4), 542–570. <https://doi.org/10.1111/ejed.12533>
- Hubbard, J. K., Potts, M. A., & Couch, B. A. (2017). How question types reveal student thinking: An experimental comparison of multiple-true-false and free-response formats. *CBE–Life Sciences Education*, 16(2). <https://doi.org/10.1187/cbe.16-12-0339>
- Jang, M., & Lukasiewicz, T. (2023). Consistency analysis of chatgpt. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. <https://doi.org/10.18653/v1/2023.emnlp-main.991>
- Krathwohl, D. R. (2002). A revision of bloom's taxonomy: An overview. *Theory into Practice*, 41(4), 212–218.
- Lan, Y.-J., & Chen, N.-S. (2024). Teachers' agency in the era of llm and generative ai: Designing pedagogical ai agents. *Educational Technology & Society*, 27(1), I–XVIII. [https://doi.org/10.30191/ETS.202401_27\(1\).PP01](https://doi.org/10.30191/ETS.202401_27(1).PP01)
- Lee, J., & Song, Y.-T. (2024). College exam grader using llm ai models. *2024 IEEE/ACIS 27th International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD)*, 282–289. <https://doi.org/10.1109/snspd61259.2024.10673924>
- Lin, D., Wen, Y., Wang, W., & Su, Y. (2024). Enhanced sentiment intensity regression through lora fine-tuning on llama 3. *IEEE Access*, 12, 108072–108087. <https://doi.org/10.1109/access.2024.3438353>
- Lu, Y., Huang, H., & Xu, R. (2021). Automatic short answer grading via multiway attention networks. *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 6270–6279.
- Månsson, D., & Norgren, M. (2016). On the assessment of written exams and possible bias due to dynamic rater effects emerging from student initials and rater fatigue. *Högre utbildning*, 6(1), 21–30. <https://doi.org/10.23865/hu.v6.751>
- Montanari, L., Cilurzo, F., Selmin, F., Conti, B., Genta, I., Poletti, G., & Valvo, L. (2003). Poly(lactide-co-glycolide) microspheres containing Bupivacaine: Comparison between gamma and beta irradiation effects. *Journal of Controlled Release*, 90(3), 281–290. [https://doi.org/10.1016/S0168-3659\(03\)00153-6](https://doi.org/10.1016/S0168-3659(03)00153-6)

- Prokop, A., Helling, H. J., Hahn, U., Udomkaewkanjana, C., & Rehm, K. E. (2005). Biodegradable implants for Pipkin fractures. *Clinical Orthopaedics and Related Research*, 12(32), 226–233.
- Riaz, A., Medard, M., Duffy, K. R., & Yazicigil, R. T. (2022). A Universal Maximum Likelihood GRAND decoder in 40nm CMOS. *2022 14th International Conference on COMmunication Systems and NETworkS (COMSNETS)*, 421–423. <https://doi.org/10.1109/COMSNETS53615.2022.9668514>
- Ridley, J., & Husband, G. (2022). Automated essay scoring: A systematic literature review. *Computers and Education: Artificial Intelligence*, 3, 100064.
- Rodrigo, A., & Penas, A. (2015). On evaluating the contribution of validation for question answering. *IEEE Transactions on Knowledge and Data Engineering*, 27(4), 1157–1161. <https://doi.org/10.1109/tkde.2014.2373363>
- Ryznar, M. (2023). Exams in the time of chatgpt. *Washington and Lee Law Review Online*, 80(5), 305–305.
- Sakai, H., & Lam, S. S. (2025, March). Quad-llm-mltc: Large language models ensemble learning for healthcare text multi-label classification [arXiv preprint]. <https://doi.org/10.48550/arXiv.2502.14189>
- Scouller, K. (1998). The influence of assessment method on students' learning approaches: Multiple choice question examination versus assignment essay. *Higher Education*, 35(4), 453–472.
- Shermis, M. D. (2014). *The state of automated essay scoring*. Routledge.
- Smith, R. (2005). *Biodegradable polymers for industrial applications*. CRC Press.
- Stanny, C. J. (2016). Reevaluating bloom's taxonomy: What measurable verbs can and cannot say. *Journal on Excellence in College Teaching*, 27(1), 1–22.
- Tekin, S. F., Ilhan, F., Huang, T., Hu, S., & Liu, L. (2024). Llm-topla: Efficient llm ensemble by maximising diversity. *Findings of the Association for Computational Linguistics: EMNLP 2024*, 11951–11966. <https://doi.org/10.18653/v1/2024.findings-emnlp.698>
- Tongsilp, A., Tangdhanakanond, K., & Chaimangkol, N. (2024). Development of automated scoring system for thai writing ability test of primary education level. *Kasetsart Journal of Social Sciences*, 45(3), 737–748. <https://doi.org/10.34044/j.kjss.2024.45.3.05>
- Vajjala, S., Alhafni, B., Bannò, S., Maurya, K. K., & Kochmar, E. (2025). Opportunities and challenges of llms in education: An nlp perspective [Preprint, submitted on 30 Jul 2025]. <https://doi.org/10.48550/arXiv.2507.22753>

Zohar, A., & Dori, Y. J. (2004). Higher order thinking skills and low-achieving students: Are they mutually exclusive? *The Journal of the Learning Sciences*, 13(2), 145–181.

APPENDICES

APPENDIX A

LIST OF HIGHLY-WEIGHTED TERMS

This is a draft of the official Latex template for SIIT Master or PhD Thesis. It has been revised in April 2022 to try to follow the SIIT requirements given in the file "ThesisTemplate_format_25Nov19.pdf" found from <https://reg.siit.tu.ac.th/registrar/requestform/ThesisTemplate25Nov19.pdf> and also in <https://reg.siit.tu.ac.th/registrar/downloadform.asp?flag=1&avs2321316=1>. Another revision was made in February 2025 to change signatory names and correct minor formatting inconsistencies. In the latter address, you can find a LaTeX template as well. This draft is the revision of that template.

A.1 Main Heading 1

Sirindhorn International Institute of Technology (SIIT) offers three master of engineering programs, namely, Master of Engineering Program in Engineering Technology, Master of Engineering Program in Information and Communication Technology for Embedded Systems, and Master of Engineering Program in Logistics and Supply Chain Systems Engineering (Chen & Winterbone, 2014).

APPENDIX B

MATLAB CODES

Here is another appendix. This is a draft of the official Latex template for SIIT Master or PhD Thesis. It has been revised in April 2022 to try to follow the SIIT requirements given in the file "ThesisTemplate_format_25Nov19.pdf" found from <https://reg.siiit.tu.ac.th/registrar/requestform/ThesisTemplate25Nov19.pdf> and also in <https://reg.siiit.tu.ac.th/registrar/downloadform.asp?flag=1&avs2321316=1>. Another revision was made in February 2025 to change signatory names and correct minor formatting inconsistencies. In the latter address, you can find a LaTeX template as well. This draft is the revision of that template.

B.1 Main Heading 1

Sirindhorn International Institute of Technology (SIIT) offers three master of engineering programs, namely, Master of Engineering Program in Engineering Technology, Master of Engineering Program in Information and Communication Technology for Embedded Systems, and Master of Engineering Program in Logistics and Supply Chain Systems Engineering (Chen & Winterbone, 2014).

BIOGRAPHY

Name	Chutipon Trirattananurak
Education	2020: Bachelor of Engineering (Electrical Engineering) Thammasat University 2024: Master of Engineering (Artificial Intelligence and Internet of Things) Sirindhorn International Institute of Technology Thammasat University

Publications

- Institute, I. , Foo, M. E. , & Gopinath, S. C. B. (2022). Feasibility of graphene in biomedical applications. to be appear in *Biomedicine & Pharmacotherapy*.
- Institute, I., Hanot, C. C., Choi, Y. S., Anani, T. B., Soundarrajan, D., & David, A. E. (2015). Effects of iron-oxide nanoparticle surface chemistry on uptake kinetics and cytotoxicity in CHO- K1 cells. *International Journal of Molecular Sciences*, 17(1), 226-230.
- Institute, I., Sinervo, A., & Arkkio, A. (2012). Modeling two-pole cage induction machine equipped with embedded force actuator. *Proceedings of 10th International Conference on Vibrations in Rotating Machinery* (pp. 765-774). New York: Woodhead Publishing.

